eda_life_style_data_v1.3

November 6, 2025

1 Life_Style_Data.csv

1.1 Chargement et inspection initiale

1.1.1 Objectif

Vérifier que le fichier `Life_Style_Data.csv` est correctement chargé et obtenir une vue d'ensemble : - dimensions du jeu de données (nombre de lignes et de colonnes), - typage des variables (numériques, catégorielles, booléennes), - détection des valeurs manquantes et doublons, - cohérence générale du corpus avant analyse statistique.

1.1.2 Interprétation attendue

- Le dataset devrait comporter une cinquantaine de colonnes, combinant données **physiques, comportementales et nutritionnelles**.
- Les types doivent être correctement inférés : numériques (`Age`, `Weight`, `Calories_Burned`), catégorielles (`Gender`, `Workout_Type`), booléennes (`is_healthy`).
- On identifiera les **colonnes à forte proportion de valeurs manquantes** afin de décider ultérieurement d'un traitement (imputation, suppression, ou encodage spécial).
- La détection de **doublons** permettra d'évaluer la fiabilité du corpus et la diversité des observations.
- Enfin, l'inspection des noms de colonnes garantira la cohérence du dictionnaire des données avant les analyses statistiques.

1.1.3 Importation des librairies essentielles

[57]:
```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import kagglehub

# === Localiser automatiquement le package 'themes' en remontant l'arborescence␣
↪===
```

```
from pathlib import Path
import sys

root = Path.cwd()           # ex: .../DuckDB/eda/life_style_data
themes_root = None

for p in [root, *root.parents]:   # remonte: life_style_data -> eda -> DuckDB
 ↪-> ...
    if (p / "themes").is_dir():
        themes_root = p
        break

if themes_root is None:
    raise FileNotFoundError(
        f"Impossible de trouver le dossier 'themes' en partant de {root}. "
        "Vérifie l'arborescence (il doit exister un dossier 'themes/' quelque
 ↪part au-dessus)."
    )

if str(themes_root) not in sys.path:
    sys.path.insert(0, str(themes_root))

print("  Ajouté au PYTHONPATH :", themes_root)
print("  Contenu de", themes_root, ":", [x.name for x in (themes_root).
 ↪iterdir()])


from themes.train_me import set_trainme_theme
```

```
  Ajouté au PYTHONPATH : c:\Users\fback\Desktop\Projets\Dev\Projet Alyra\DuckDB
  Contenu de c:\Users\fback\Desktop\Projets\Dev\Projet Alyra\DuckDB : ['.conda',
'.vscode', '.~lock.Exploration des données - v1.1.odt#', 'datasets',
'duckdb.exe', 'eda', 'Exploration des données - v1.0.odt', 'Exploration des
données - v1.0.pdf', 'Exploration des données - v1.1.odt', 'themes']
```

1.1.4 Configuration d'affichage pour plus de lisibilité

[58]:
```
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 120)
```

1.1.5 Thème "TrAIn.me"

Il applique un fond dark propre, une grille discrète, des ticks lisibles, et une palette bleus/cians cohérente.

[59]:
```
# ==============================
# Thème TrAIn.me (Seaborn/Matplotlib)
# ==============================
```

```
# Booléen de contrôle : True = thème TrAIn.me (dark), False = style clair␣
 ↪"ticks"
USE_DARK_THEME = True  #  modifie ici selon le contexte

# Activer le thème au début du notebook :
palette_trainme = set_trainme_theme(context="talk", font_scale=1.05,␣
 ↪use_dark=True)
palette_trainme
```

[59]: ['#00E5FF', '#00BFFF', '#0A84FF', '#1E90FF', '#00FFFF', '#64D8FF']

1.1.6 Chargement du dataset

[60]:
```
# Construction du chemin relatif
# file_path = "../../datasets/Life_Style_Data.csv"

# Téléchargement du dataset depuis Kaggle
dataset_path = kagglehub.dataset_download("jockeroika/life-style-data")

# Recherche du fichier CSV dans le dossier téléchargé
csv_files = [f for f in os.listdir(dataset_path) if f.endswith(".csv")]
print("Fichiers trouvés :", csv_files)

# Construction du chemin complet vers le CSV
file_path = os.path.join(dataset_path, csv_files[0])


# Lecture du fichier CSV
df = pd.read_csv(file_path)

# Informations de confirmation
print("Dataset chargé avec succès !")
print(f"Dimensions : {df.shape[0]} lignes et {df.shape[1]} colonnes\n")
```

```
Warning: Looks like you're using an outdated `kagglehub` version, please
consider updating (latest version: 0.3.13)
Fichiers trouvés : ['Final_data.csv', 'meal_metadata.csv']
Dataset chargé avec succès !
Dimensions : 20000 lignes et 54 colonnes
```

1.1.7 Aperçu général des premières lignes

[61]:
```
display(df.head())
```

```
    Age  Gender  Weight (kg)  Height (m)  Max_BPM  Avg_BPM  Resting_BPM ␣
 ↪Session_Duration (hours)  Calories_Burned  \
```

```
0  34.91    Male        65.27        1.62   188.58   157.65        69.05
  ↪             1.00          1080.90
1  23.37  Female        56.41        1.55   179.43   131.75        73.18
  ↪             1.37          1809.91
2  33.20  Female        58.98        1.67   175.04   123.95        54.96
  ↪             0.91           802.26
3  38.69  Female        93.78        1.70   191.21   155.10        50.07
  ↪             1.10          1450.79
4  45.09    Male        52.42        1.88   193.58   152.88        70.84
  ↪             1.08          1166.40

  Workout_Type  Fat_Percentage  Water_Intake (liters)  Workout_Frequency (days/
  ↪week)  Experience_Level    BMI  \
0     Strength       26.800377                   1.50
  ↪3.99              2.01  24.87
1         HIIT       27.655021                   1.90
  ↪4.00              2.01  23.48
2       Cardio       24.320821                   1.88
  ↪2.99              1.02  21.15
3         HIIT       32.813572                   2.50
  ↪3.99              1.99  32.45
4     Strength       17.307319                   2.91
  ↪4.00              2.00  14.83

   Daily meals frequency  Physical exercise   Carbs  Proteins   Fats  Calories
  ↪meal_name  meal_type   diet_type  \
0                   2.99               0.01  267.68    106.05  71.63    1806.0
  ↪  Other      Lunch       Vegan
1                   3.01               0.97  214.32     85.41  56.97    1577.0
  ↪  Other      Lunch  Vegetarian
2                   1.99              -0.02  246.04     98.11  65.48    1608.0
  ↪  Other  Breakfast       Paleo
3                   3.00               0.04  203.22     80.84  54.56    2657.0
  ↪  Other      Lunch       Paleo
4                   3.00               3.00  332.79    133.05  88.43    1470.0
  ↪  Other  Breakfast       Vegan

   sugar_g  sodium_mg  cholesterol_mg  serving_size_g cooking_method
  ↪prep_time_min  cook_time_min  rating  \
0    31.77    1729.94          285.05          120.47        Grilled          16.
  ↪24         110.79    1.31
1    12.34     693.08          300.61          109.15          Fried          16.
  ↪47          12.01    1.92
2    42.81    2142.48          215.42          399.43         Boiled          54.
  ↪35           6.09    4.70
```

```
3     9.34     123.20            9.70          314.31          Fried          27.
↪73         103.72    4.85
4    23.78    1935.11          116.89           99.22          Baked          34.
↪16          46.55    3.07

    Name of Exercise  Sets   Reps                                              ␣
↪Benefit  Burns Calories (per 30 min)  \
0   Decline Push-ups  4.99  20.91             Improves shoulder health and␣
↪posture                        342.58
1        Bear Crawls  4.01  16.15                          Strengthens lower␣
↪abs                          357.16
2               Dips  5.00  21.90                          Builds chest␣
↪strength                       359.63
3  Mountain Climbers  4.01  16.92  Improves coordination and cardiovascular␣
↪health                         351.65
4        Bicep Curls  4.99  15.01      Targets obliques and improves core␣
↪rotation                       329.36

     Target Muscle Group        Equipment Needed Difficulty Level  Body Part␣
↪Type of Muscle                  Workout  \
0     Shoulders, Triceps           Cable Machine         Advanced       Legs    ␣
↪      Lats          Dumbbell flyes
1  Back, Core, Shoulders             Step or Box     Intermediate      Chest    ␣
↪      Lats          Lateral raises
2     Quadriceps, Glutes             Step or Box     Intermediate       Arms ␣
↪Grip Strength    Standing calf raises
3       Biceps, Forearms  Parallel Bars or Chair         Advanced  Shoulders    ␣
↪     Upper  Incline dumbbell flyes
4         Chest, Triceps                    Wall         Advanced        Abs ␣
↪Wrist Flexors          Military press

    BMI_calc  cal_from_macros  pct_carbs  protein_per_kg   pct_HRR  pct_maxHR ␣
↪cal_balance  lean_mass_kg  \
0  24.870447          2139.59   0.500432        1.624789  0.741237   0.835985   ␣
↪   725.10     47.777394
1  23.479709          1711.65   0.500850        1.514093  0.551247   0.734270   ␣
↪  -232.91     40.809803
2  21.148123          1965.92   0.500610        1.663445  0.574534   0.708124   ␣
↪   805.74     44.635580
3  32.449827          1627.28   0.499533        0.862017  0.744155   0.811150   ␣
↪  1206.21     63.007432
4  14.831372          2659.23   0.500581        2.538153  0.668405   0.789751   ␣
↪   303.60     43.347504

   expected_burn  Burns Calories (per 30 min)_bc Burns_Calories_Bin
0       685.1600                   7.260425e+19             Medium
```

```
1       978.6184                    1.020506e+20              High
2       654.5266                    1.079607e+20              High
3       773.6300                    8.987921e+19              High
4       711.4176                    5.264685e+19               Low
```

1.1.8 Informations générales sur le typage et la complétude

[62]:
```
df_info = df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 54 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Age                             20000 non-null  float64
 1   Gender                          20000 non-null  object
 2   Weight (kg)                     20000 non-null  float64
 3   Height (m)                      20000 non-null  float64
 4   Max_BPM                         20000 non-null  float64
 5   Avg_BPM                         20000 non-null  float64
 6   Resting_BPM                     20000 non-null  float64
 7   Session_Duration (hours)        20000 non-null  float64
 8   Calories_Burned                 20000 non-null  float64
 9   Workout_Type                    20000 non-null  object
 10  Fat_Percentage                  20000 non-null  float64
 11  Water_Intake (liters)           20000 non-null  float64
 12  Workout_Frequency (days/week)   20000 non-null  float64
 13  Experience_Level                20000 non-null  float64
 14  BMI                             20000 non-null  float64
 15  Daily meals frequency           20000 non-null  float64
 16  Physical exercise               20000 non-null  float64
 17  Carbs                           20000 non-null  float64
 18  Proteins                        20000 non-null  float64
 19  Fats                            20000 non-null  float64
 20  Calories                        20000 non-null  float64
 21  meal_name                       20000 non-null  object
 22  meal_type                       20000 non-null  object
 23  diet_type                       20000 non-null  object
 24  sugar_g                         20000 non-null  float64
 25  sodium_mg                       20000 non-null  float64
 26  cholesterol_mg                  20000 non-null  float64
 27  serving_size_g                  20000 non-null  float64
 28  cooking_method                  20000 non-null  object
 29  prep_time_min                   20000 non-null  float64
 30  cook_time_min                   20000 non-null  float64
 31  rating                          20000 non-null  float64
 32  Name of Exercise                20000 non-null  object
 33  Sets                            20000 non-null  float64
```

```
 34  Reps                              20000 non-null  float64
 35  Benefit                           20000 non-null  object
 36  Burns Calories (per 30 min)       20000 non-null  float64
 37  Target Muscle Group               20000 non-null  object
 38  Equipment Needed                  20000 non-null  object
 39  Difficulty Level                  20000 non-null  object
 40  Body Part                         20000 non-null  object
 41  Type of Muscle                    20000 non-null  object
 42  Workout                           20000 non-null  object
 43  BMI_calc                          20000 non-null  float64
 44  cal_from_macros                   20000 non-null  float64
 45  pct_carbs                         20000 non-null  float64
 46  protein_per_kg                    20000 non-null  float64
 47  pct_HRR                           20000 non-null  float64
 48  pct_maxHR                         20000 non-null  float64
 49  cal_balance                       20000 non-null  float64
 50  lean_mass_kg                      20000 non-null  float64
 51  expected_burn                     20000 non-null  float64
 52  Burns Calories (per 30 min)_bc    20000 non-null  float64
 53  Burns_Calories_Bin                20000 non-null  object
dtypes: float64(39), object(15)
memory usage: 8.2+ MB
```

1.1.9 Statistiques descriptives des variables numériques

[63]:
```
display(df.describe().T)
```

```
                                count          mean           std           ␣
↪min           25%           50%  \
Age                             20000.0  3.885145e+01  1.211458e+01  1.
↪800000e+01  2.817000e+01  3.986500e+01
Weight (kg)                     20000.0  7.389883e+01  2.117301e+01  3.
↪918000e+01  5.816000e+01  7.000000e+01
Height (m)                      20000.0  1.723094e+00  1.270328e-01  1.
↪490000e+00  1.620000e+00  1.710000e+00
Max_BPM                         20000.0  1.798897e+02  1.151081e+01  1.
↪593100e+02  1.700575e+02  1.801400e+02
Avg_BPM                         20000.0  1.437043e+02  1.426769e+01  1.
↪190700e+02  1.312200e+02  1.429900e+02
Resting_BPM                     20000.0  6.219581e+01  7.289351e+00  4.
↪949000e+01  5.596000e+01  6.220000e+01
Session_Duration (hours)        20000.0  1.259446e+00  3.413362e-01  4.
↪900000e-01  1.050000e+00  1.270000e+00
Calories_Burned                 20000.0  1.280110e+03  5.022290e+02  3.
↪231100e+02  9.108000e+02  1.231450e+03
Fat_Percentage                  20000.0  2.610131e+01  4.996160e+00  1.
↪133313e+01  2.238781e+01  2.582250e+01
```

```
Water_Intake (liters)           20000.0  2.627486e+00  6.047240e-01  1.
 ↪460000e+00  2.170000e+00  2.610000e+00
Workout_Frequency (days/week)   20000.0  3.318629e+00  9.111979e-01  1.
 ↪940000e+00  2.980000e+00  3.010000e+00
Experience_Level                20000.0  1.809176e+00  7.362036e-01  1.
 ↪000000e+00  1.010000e+00  1.990000e+00
BMI                             20000.0  2.492168e+01  6.701126e+00  1.
 ↪204000e+01  2.009750e+01  2.412000e+01
Daily meals frequency           20000.0  2.864804e+00  6.366486e-01  1.
 ↪950000e+00  2.020000e+00  3.000000e+00
Physical exercise               20000.0  4.525030e-01  9.866276e-01 -7.
 ↪000000e-02 -1.000000e-02  1.000000e-02
Carbs                           20000.0  2.497786e+02  5.510862e+01  1.
 ↪377200e+02  2.075475e+02  2.429000e+02
Proteins                        20000.0  9.991829e+01  2.204367e+01  5.
 ↪517000e+01  8.304000e+01  9.714500e+01
Fats                            20000.0  6.661217e+01  1.469928e+01  3.
 ↪659000e+01  5.534000e+01  6.477000e+01
Calories                        20000.0  2.024418e+03  5.418915e+02  7.
 ↪810000e+02  1.634000e+03  1.919000e+03
sugar_g                         20000.0  2.453104e+01  1.444610e+01 -6.
 ↪700000e-01  1.199000e+01  2.406000e+01
sodium_mg                       20000.0  1.258140e+03  7.166190e+02  1.
 ↪922000e+01  6.451275e+02  1.267650e+03
cholesterol_mg                  20000.0  1.484656e+02  8.738282e+01 -3.
 ↪890000e+00  7.197000e+01  1.497950e+02
serving_size_g                  20000.0  3.027195e+02  1.157119e+02  9.
 ↪595000e+01  2.072175e+02  3.000150e+02
prep_time_min                   20000.0  3.351745e+01  1.652687e+01  3.
 ↪950000e+00  1.858750e+01  3.433000e+01
cook_time_min                   20000.0  6.150216e+01  3.357975e+01  3.
 ↪350000e+00  3.247750e+01  6.092000e+01
rating                          20000.0  2.989303e+00  1.157692e+00  9.
 ↪300000e-01  1.940000e+00  3.000000e+00
Sets                            20000.0  4.425059e+00  5.795959e-01  2.
 ↪960000e+00  4.000000e+00  4.020000e+00
Reps                            20000.0  1.942732e+01  3.679707e+00  4.
 ↪850000e+00  1.612000e+01  1.990000e+01
Burns Calories (per 30 min)     20000.0  3.440465e+02  3.213878e+01  1.
 ↪287500e+02  3.376000e+02  3.486050e+02
BMI_calc                        20000.0  2.492165e+01  6.701144e+00  1.
 ↪203791e+01  2.009497e+01  2.411910e+01
cal_from_macros                 20000.0  1.998297e+03  4.408484e+02  1.
 ↪105570e+03  1.661023e+03  1.943130e+03
pct_carbs                       20000.0  4.999830e-01  1.455490e-03  4.
 ↪924342e-01  4.990538e-01  4.999810e-01
```

```
protein_per_kg                  20000.0  1.460142e+00  5.189455e-01  5.
 ↪167064e-01  1.076294e+00  1.382260e+00
pct_HRR                         20000.0  6.990053e-01  1.448804e-01  3.
 ↪713439e-01  5.836562e-01  6.862837e-01
pct_maxHR                       20000.0  8.023050e-01  9.661268e-02  5.
 ↪997889e-01  7.276759e-01  7.948340e-01
cal_balance                     20000.0  7.443087e+02  7.209466e+02 -1.
 ↪266220e+03  2.614325e+02  6.911900e+02
lean_mass_kg                    20000.0  5.378638e+01  1.249874e+01  3.
 ↪094626e+01  4.458704e+01  5.120491e+01
expected_burn                   20000.0  8.663523e+02  2.503171e+02  2.
 ↪198528e+02  7.140983e+02  8.687214e+02
Burns Calories (per 30 min)_bc  20000.0  8.631802e+19  3.197579e+19  2.
 ↪491905e+16  6.441978e+19  8.371578e+19
```

	75%	max
Age	4.963000e+01	5.967000e+01
Weight (kg)	8.610000e+01	1.307700e+02
Height (m)	1.800000e+00	2.010000e+00
Max_BPM	1.894250e+02	1.996400e+02
Avg_BPM	1.560600e+02	1.698400e+02
Resting_BPM	6.809000e+01	7.450000e+01
Session_Duration (hours)	1.460000e+00	2.020000e+00
Calories_Burned	1.553112e+03	2.890820e+03
Fat_Percentage	2.967603e+01	3.500000e+01
Water_Intake (liters)	3.120000e+00	3.730000e+00
Workout_Frequency (days/week)	4.000000e+00	5.060000e+00
Experience_Level	2.020000e+00	3.050000e+00
BMI	2.856000e+01	5.023000e+01
Daily meals frequency	3.010000e+00	4.040000e+00
Physical exercise	4.000000e-02	4.050000e+00
Carbs	2.839750e+02	4.624900e+02
Proteins	1.136400e+02	1.853400e+02
Fats	7.575250e+01	1.234200e+02
Calories	2.360000e+03	3.641000e+03
sugar_g	3.749000e+01	5.051000e+01
sodium_mg	1.850893e+03	2.527270e+03
cholesterol_mg	2.218400e+02	3.039900e+02
serving_size_g	4.018600e+02	5.075200e+02
prep_time_min	4.794000e+01	6.129000e+01
cook_time_min	8.937500e+01	1.214600e+02
rating	4.000000e+00	5.060000e+00
Sets	5.000000e+00	5.050000e+00
Reps	2.288000e+01	3.012000e+01
Burns Calories (per 30 min)	3.604725e+02	3.817100e+02
BMI_calc	2.856262e+01	5.022954e+01
cal_from_macros	2.271950e+03	3.699540e+03

```
pct_carbs                        5.009103e-01  5.078886e-01
protein_per_kg                   1.750495e+00  3.916881e+00
pct_HRR                          7.981957e-01  1.073939e+00
pct_maxHR                        8.692108e-01  1.047032e+00
cal_balance                      1.176290e+03  3.075580e+03
lean_mass_kg                     6.193902e+01  9.011737e+01
expected_burn                    1.012533e+03  1.477109e+03
Burns Calories (per 30 min)_bc   1.100442e+20  1.756614e+20
```

1.1.10 Analyse des valeurs manquantes et doublons

[64]:
```
missing_values = df.isnull().sum().sort_values(ascending=False)
duplicates = df.duplicated().sum()

print("\n--- Valeurs manquantes par colonne (top 10) ---")
display(missing_values.head(10))

print(f"\nNombre total de doublons détectés : {duplicates}")
```

```
--- Valeurs manquantes par colonne (top 10) ---

Age                              0
Body Part                        0
prep_time_min                    0
cook_time_min                    0
rating                           0
Name of Exercise                 0
Sets                             0
Reps                             0
Benefit                          0
Burns Calories (per 30 min)      0
dtype: int64


Nombre total de doublons détectés : 0
```

1.1.11 Liste complète des colonnes disponibles

[65]:
```
print("\n--- Liste des colonnes du dataset ---")
print(df.columns.tolist())
```

```
--- Liste des colonnes du dataset ---
['Age', 'Gender', 'Weight (kg)', 'Height (m)', 'Max_BPM', 'Avg_BPM',
'Resting_BPM', 'Session_Duration (hours)', 'Calories_Burned', 'Workout_Type',
'Fat_Percentage', 'Water_Intake (liters)', 'Workout_Frequency (days/week)',
'Experience_Level', 'BMI', 'Daily meals frequency', 'Physical exercise',
'Carbs', 'Proteins', 'Fats', 'Calories', 'meal_name', 'meal_type', 'diet_type',
```

```
'sugar_g', 'sodium_mg', 'cholesterol_mg', 'serving_size_g', 'cooking_method',
'prep_time_min', 'cook_time_min', 'rating', 'Name of Exercise', 'Sets', 'Reps',
'Benefit', 'Burns Calories (per 30 min)', 'Target Muscle Group', 'Equipment
Needed', 'Difficulty Level', 'Body Part', 'Type of Muscle', 'Workout',
'BMI_calc', 'cal_from_macros', 'pct_carbs', 'protein_per_kg', 'pct_HRR',
'pct_maxHR', 'cal_balance', 'lean_mass_kg', 'expected_burn', 'Burns Calories
(per 30 min)_bc', 'Burns_Calories_Bin']
```

1.2 Statistiques descriptives et distribution des variables

1.2.1 Objectif

Explorer la distribution des variables numériques afin de comprendre la variabilité du dataset, identifier d'éventuelles anomalies et détecter les premiers signaux statistiques utiles.

1.2.2 Interprétation attendue

- Confirmer la cohérence statistique des variables physiologiques (âge, poids, taille, IMC).
- Vérifier la normalité ou l'asymétrie de certaines distributions (Calories_Burned, BMI…).
- Identifier visuellement les outliers potentiels pour un traitement futur.

1.2.3 Activation du thème TrAIn.me

[66]:
```
if USE_DARK_THEME:
    palette_trainme = set_trainme_theme(context="talk", font_scale=1.05,␣
 ↪use_dark=True)
else:
    sns.set_style("ticks")              # style clair sobre, axes renforcés
    sns.set_context("talk", font_scale=1.05)
    palette_trainme = sns.color_palette("deep")  # palette par défaut sobre
    print("  Style activé : 'ticks' → sobre, axes renforcés (présentation pro)")

# Exemple de vérification :
print("Palette active :", palette_trainme.as_hex() if hasattr(palette_trainme,␣
 ↪"as_hex") else palette_trainme)
```

```
Palette active : ['#00E5FF', '#00BFFF', '#0A84FF', '#1E90FF', '#00FFFF',
'#64D8FF']
```

1.2.4 Statistiques descriptives globales

[67]:
```
stats = df.describe().T
display(stats)
# Vérification des bornes extrêmes sur les principales métriques
print(f"Âge moyen : {df['Age'].mean():.1f} ans")
print(f"Poids moyen : {df['Weight (kg)'].mean():.1f} kg")
print(f"Calories moyennes brûlées : {df['Calories_Burned'].mean():.0f} kcal")
```

```
print(f"IMC moyen : {df['BMI'].mean():.1f}")
```

```
                                 count          mean           std
↪min           25%           50%  \
Age                              20000.0  3.885145e+01  1.211458e+01  1.
↪800000e+01  2.817000e+01  3.986500e+01
Weight (kg)                      20000.0  7.389883e+01  2.117301e+01  3.
↪918000e+01  5.816000e+01  7.000000e+01
Height (m)                       20000.0  1.723094e+00  1.270328e-01  1.
↪490000e+00  1.620000e+00  1.710000e+00
Max_BPM                          20000.0  1.798897e+02  1.151081e+01  1.
↪593100e+02  1.700575e+02  1.801400e+02
Avg_BPM                          20000.0  1.437043e+02  1.426769e+01  1.
↪190700e+02  1.312200e+02  1.429900e+02
Resting_BPM                      20000.0  6.219581e+01  7.289351e+00  4.
↪949000e+01  5.596000e+01  6.220000e+01
Session_Duration (hours)         20000.0  1.259446e+00  3.413362e-01  4.
↪900000e-01  1.050000e+00  1.270000e+00
Calories_Burned                  20000.0  1.280110e+03  5.022290e+02  3.
↪231100e+02  9.108000e+02  1.231450e+03
Fat_Percentage                   20000.0  2.610131e+01  4.996160e+00  1.
↪133313e+01  2.238781e+01  2.582250e+01
Water_Intake (liters)            20000.0  2.627486e+00  6.047240e-01  1.
↪460000e+00  2.170000e+00  2.610000e+00
Workout_Frequency (days/week)    20000.0  3.318629e+00  9.111979e-01  1.
↪940000e+00  2.980000e+00  3.010000e+00
Experience_Level                 20000.0  1.809176e+00  7.362036e-01  1.
↪000000e+00  1.010000e+00  1.990000e+00
BMI                              20000.0  2.492168e+01  6.701126e+00  1.
↪204000e+01  2.009750e+01  2.412000e+01
Daily meals frequency            20000.0  2.864804e+00  6.366486e-01  1.
↪950000e+00  2.020000e+00  3.000000e+00
Physical exercise                20000.0  4.525030e-01  9.866276e-01 -7.
↪000000e-02 -1.000000e-02  1.000000e-02
Carbs                            20000.0  2.497786e+02  5.510862e+01  1.
↪377200e+02  2.075475e+02  2.429000e+02
Proteins                         20000.0  9.991829e+01  2.204367e+01  5.
↪517000e+01  8.304000e+01  9.714500e+01
Fats                             20000.0  6.661217e+01  1.469928e+01  3.
↪659000e+01  5.534000e+01  6.477000e+01
Calories                         20000.0  2.024418e+03  5.418915e+02  7.
↪810000e+02  1.634000e+03  1.919000e+03
sugar_g                          20000.0  2.453104e+01  1.444610e+01 -6.
↪700000e-01  1.199000e+01  2.406000e+01
sodium_mg                        20000.0  1.258140e+03  7.166190e+02  1.
↪922000e+01  6.451275e+02  1.267650e+03
```

```
cholesterol_mg                  20000.0  1.484656e+02  8.738282e+01 -3.
 ↪890000e+00  7.197000e+01  1.497950e+02
serving_size_g                  20000.0  3.027195e+02  1.157119e+02  9.
 ↪595000e+01  2.072175e+02  3.000150e+02
prep_time_min                   20000.0  3.351745e+01  1.652687e+01  3.
 ↪950000e+00  1.858750e+01  3.433000e+01
cook_time_min                   20000.0  6.150216e+01  3.357975e+01  3.
 ↪350000e+00  3.247750e+01  6.092000e+01
rating                          20000.0  2.989303e+00  1.157692e+00  9.
 ↪300000e-01  1.940000e+00  3.000000e+00
Sets                            20000.0  4.425059e+00  5.795959e-01  2.
 ↪960000e+00  4.000000e+00  4.020000e+00
Reps                            20000.0  1.942732e+01  3.679707e+00  4.
 ↪850000e+00  1.612000e+01  1.990000e+01
Burns Calories (per 30 min)     20000.0  3.440465e+02  3.213878e+01  1.
 ↪287500e+02  3.376000e+02  3.486050e+02
BMI_calc                        20000.0  2.492165e+01  6.701144e+00  1.
 ↪203791e+01  2.009497e+01  2.411910e+01
cal_from_macros                 20000.0  1.998297e+03  4.408484e+02  1.
 ↪105570e+03  1.661023e+03  1.943130e+03
pct_carbs                       20000.0  4.999830e-01  1.455490e-03  4.
 ↪924342e-01  4.990538e-01  4.999810e-01
protein_per_kg                  20000.0  1.460142e+00  5.189455e-01  5.
 ↪167064e-01  1.076294e+00  1.382260e+00
pct_HRR                         20000.0  6.990053e-01  1.448804e-01  3.
 ↪713439e-01  5.836562e-01  6.862837e-01
pct_maxHR                       20000.0  8.023050e-01  9.661268e-02  5.
 ↪997889e-01  7.276759e-01  7.948340e-01
cal_balance                     20000.0  7.443087e+02  7.209466e+02 -1.
 ↪266220e+03  2.614325e+02  6.911900e+02
lean_mass_kg                    20000.0  5.378638e+01  1.249874e+01  3.
 ↪094626e+01  4.458704e+01  5.120491e+01
expected_burn                   20000.0  8.663523e+02  2.503171e+02  2.
 ↪198528e+02  7.140983e+02  8.687214e+02
Burns Calories (per 30 min)_bc  20000.0  8.631802e+19  3.197579e+19  2.
 ↪491905e+16  6.441978e+19  8.371578e+19
```

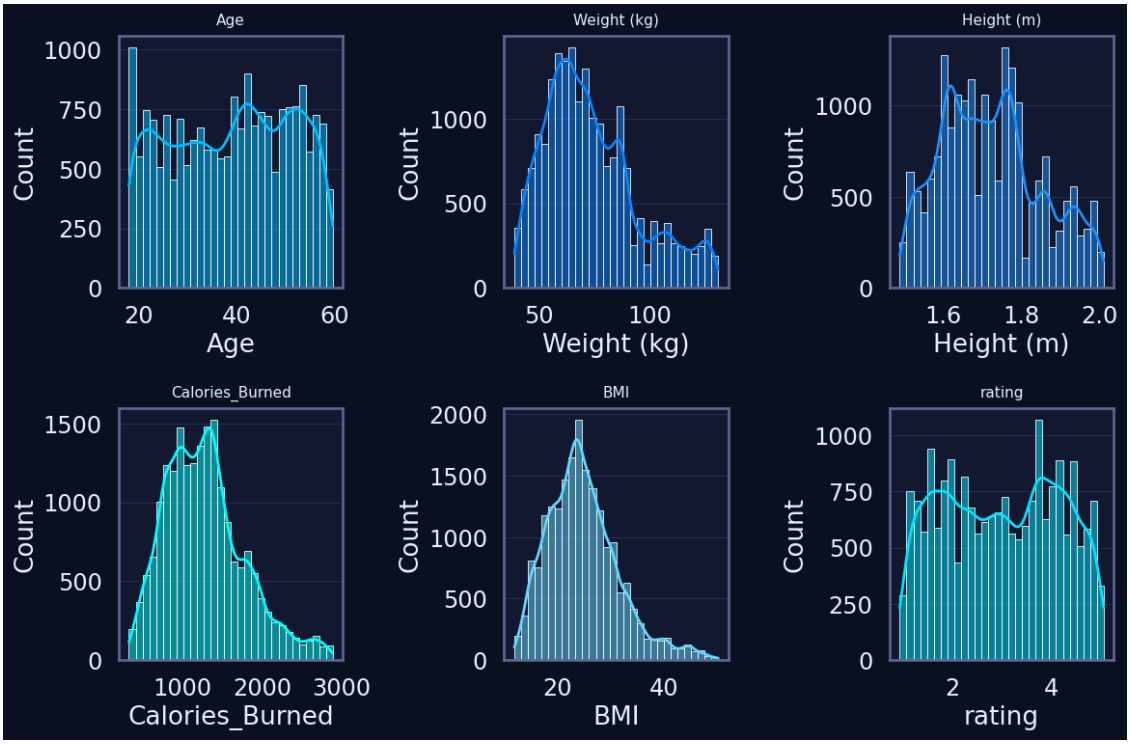
```
                                      75%           max
Age                             4.963000e+01  5.967000e+01
Weight (kg)                     8.610000e+01  1.307700e+02
Height (m)                      1.800000e+00  2.010000e+00
Max_BPM                         1.894250e+02  1.996400e+02
Avg_BPM                         1.560600e+02  1.698400e+02
Resting_BPM                     6.809000e+01  7.450000e+01
Session_Duration (hours)        1.460000e+00  2.020000e+00
Calories_Burned                 1.553112e+03  2.890820e+03
```

```
Fat_Percentage                  2.967603e+01  3.500000e+01
Water_Intake (liters)           3.120000e+00  3.730000e+00
Workout_Frequency (days/week)   4.000000e+00  5.060000e+00
Experience_Level                2.020000e+00  3.050000e+00
BMI                             2.856000e+01  5.023000e+01
Daily meals frequency           3.010000e+00  4.040000e+00
Physical exercise               4.000000e-02  4.050000e+00
Carbs                           2.839750e+02  4.624900e+02
Proteins                        1.136400e+02  1.853400e+02
Fats                            7.575250e+01  1.234200e+02
Calories                        2.360000e+03  3.641000e+03
sugar_g                         3.749000e+01  5.051000e+01
sodium_mg                       1.850893e+03  2.527270e+03
cholesterol_mg                  2.218400e+02  3.039900e+02
serving_size_g                  4.018600e+02  5.075200e+02
prep_time_min                   4.794000e+01  6.129000e+01
cook_time_min                   8.937500e+01  1.214600e+02
rating                          4.000000e+00  5.060000e+00
Sets                            5.000000e+00  5.050000e+00
Reps                            2.288000e+01  3.012000e+01
Burns Calories (per 30 min)     3.604725e+02  3.817100e+02
BMI_calc                        2.856262e+01  5.022954e+01
cal_from_macros                 2.271950e+03  3.699540e+03
pct_carbs                       5.009103e-01  5.078886e-01
protein_per_kg                  1.750495e+00  3.916881e+00
pct_HRR                         7.981957e-01  1.073939e+00
pct_maxHR                       8.692108e-01  1.047032e+00
cal_balance                     1.176290e+03  3.075580e+03
lean_mass_kg                    6.193902e+01  9.011737e+01
expected_burn                   1.012533e+03  1.477109e+03
Burns Calories (per 30 min)_bc  1.100442e+20  1.756614e+20

Âge moyen : 38.9 ans
Poids moyen : 73.9 kg
Calories moyennes brûlées : 1280 kcal
IMC moyen : 24.9
```

1.2.5 Distribution des variables clés

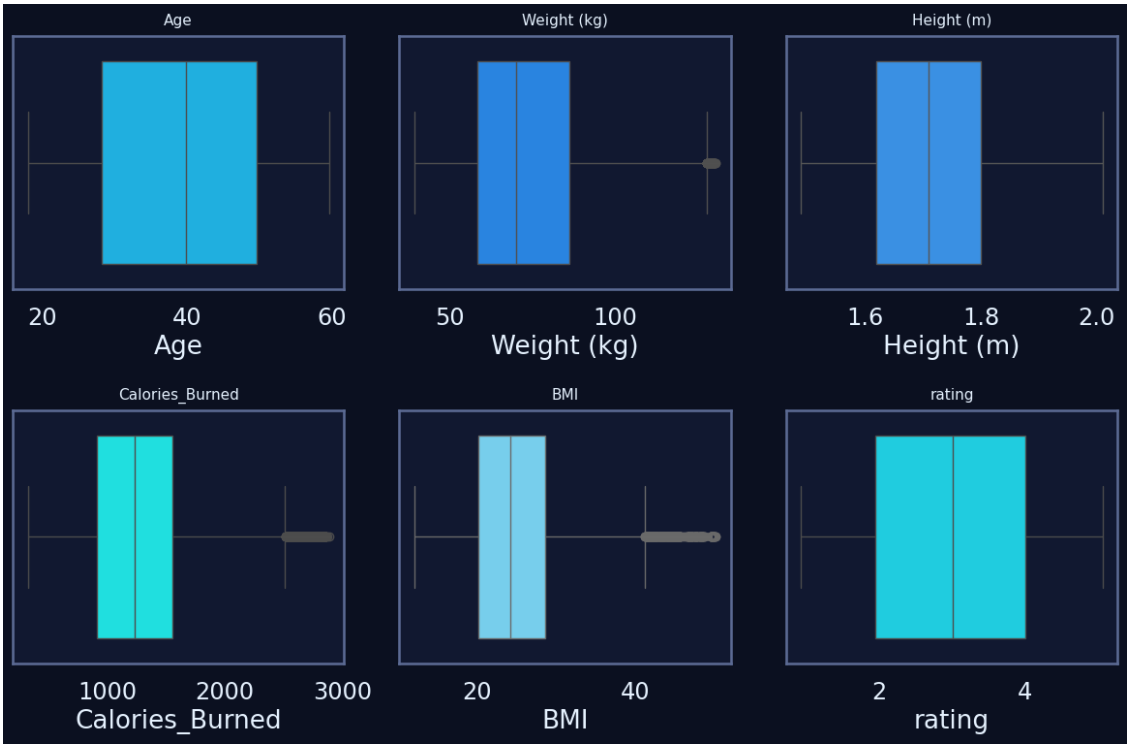
[68]:
```
num_vars = ["Age", "Weight (kg)", "Height (m)", "Calories_Burned", "BMI",␣
 ↪"rating"]

plt.figure(figsize=(12, 8))
for i, col in enumerate(num_vars, 1):
    plt.subplot(2, 3, i)
    sns.histplot(df[col], bins=30, kde=True, color=palette_trainme[i %␣
 ↪len(palette_trainme)])
    plt.title(col, fontsize=11, pad=8)
```

```
plt.tight_layout()
plt.show()
```



1.2.6 Boxplots pour repérer les outliers

[69]:
```
plt.figure(figsize=(12, 8))
for i, col in enumerate(num_vars, 1):
    plt.subplot(2, 3, i)
    sns.boxplot(x=df[col], color=palette_trainme[i % len(palette_trainme)])
    plt.title(col, fontsize=11, pad=8)
plt.tight_layout()
plt.show()
```



1.2.7 Statistiques descriptives enrichies

[73]:
```
# --- Colonnes numériques (on exclut les colonnes dérivées à magnitude extrême) ---
num_cols = df.select_dtypes(include="number").columns.tolist()
to_exclude = ["Burns Calories (per 30 min)_bc"]  # valeurs ~1e19-1e20, non comparables
num_cols = [c for c in num_cols if c not in to_exclude]

print(f"{len(num_cols)} variables numériques analysées.")

desc = df[num_cols].describe().T
desc["missing_%"] = df[num_cols].isna().mean() * 100
desc["zeros_%"]   = (df[num_cols] == 0).mean() * 100
desc["skew"]      = df[num_cols].skew()
desc["kurt"]      = df[num_cols].kurt()

display(desc.sort_values("std", ascending=False).head(12))  # top 12 par variance
```

```
38 variables numériques analysées.
                                count         mean          std         min  ↪
↪25%         50%         75%          max  \
```

```
cal_balance                  20000.0   744.308699  720.946619 -1266.2200   261.
↪43250   691.1900  1176.2900  3075.5800
sodium_mg                    20000.0  1258.139709  716.618987    19.2200   645.
↪12750  1267.6500  1850.8925  2527.2700
Calories                     20000.0  2024.418300  541.891521   781.0000  1634.
↪00000  1919.0000  2360.0000  3641.0000
Calories_Burned              20000.0  1280.109601  502.228982   323.1100   910.
↪80000  1231.4500  1553.1125  2890.8200
cal_from_macros              20000.0  1998.297076  440.848408  1105.5700  1661.
↪02250  1943.1300  2271.9500  3699.5400
expected_burn                20000.0   866.352318  250.317069   219.8528   714.
↪09825   868.7214  1012.5327  1477.1088
serving_size_g               20000.0   302.719499  115.711949    95.9500   207.
↪21750   300.0150   401.8600   507.5200
cholesterol_mg               20000.0   148.465602   87.382817    -3.8900    71.
↪97000   149.7950   221.8400   303.9900
Carbs                        20000.0   249.778592   55.108623   137.7200   207.
↪54750   242.9000   283.9750   462.4900
cook_time_min                20000.0    61.502164   33.579746     3.3500    32.
↪47750    60.9200    89.3750   121.4600
Burns Calories (per 30 min)  20000.0   344.046514   32.138782   128.7500   337.
↪60000   348.6050   360.4725   381.7100
Proteins                     20000.0    99.918290   22.043670    55.1700    83.
↪04000    97.1450   113.6400   185.3400
```

```
                             missing_%  zeros_%      skew       kurt
cal_balance                        0.0      0.0  0.302899  -0.012443
sodium_mg                          0.0      0.0  0.007964  -1.183316
Calories                           0.0      0.0  0.653412  -0.139893
Calories_Burned                    0.0      0.0  0.682426   0.276265
cal_from_macros                    0.0      0.0  0.790512   0.646217
expected_burn                      0.0      0.0  0.054182  -0.398604
serving_size_g                     0.0      0.0 -0.003414  -1.191050
cholesterol_mg                     0.0      0.0 -0.020933  -1.158937
Carbs                              0.0      0.0  0.790432   0.645807
cook_time_min                      0.0      0.0  0.059301  -1.200804
Burns Calories (per 30 min)        0.0      0.0 -3.900865  17.915841
Proteins                           0.0      0.0  0.790340   0.646705
```

1.2.8 Détection rapide d'outliers (IQR)

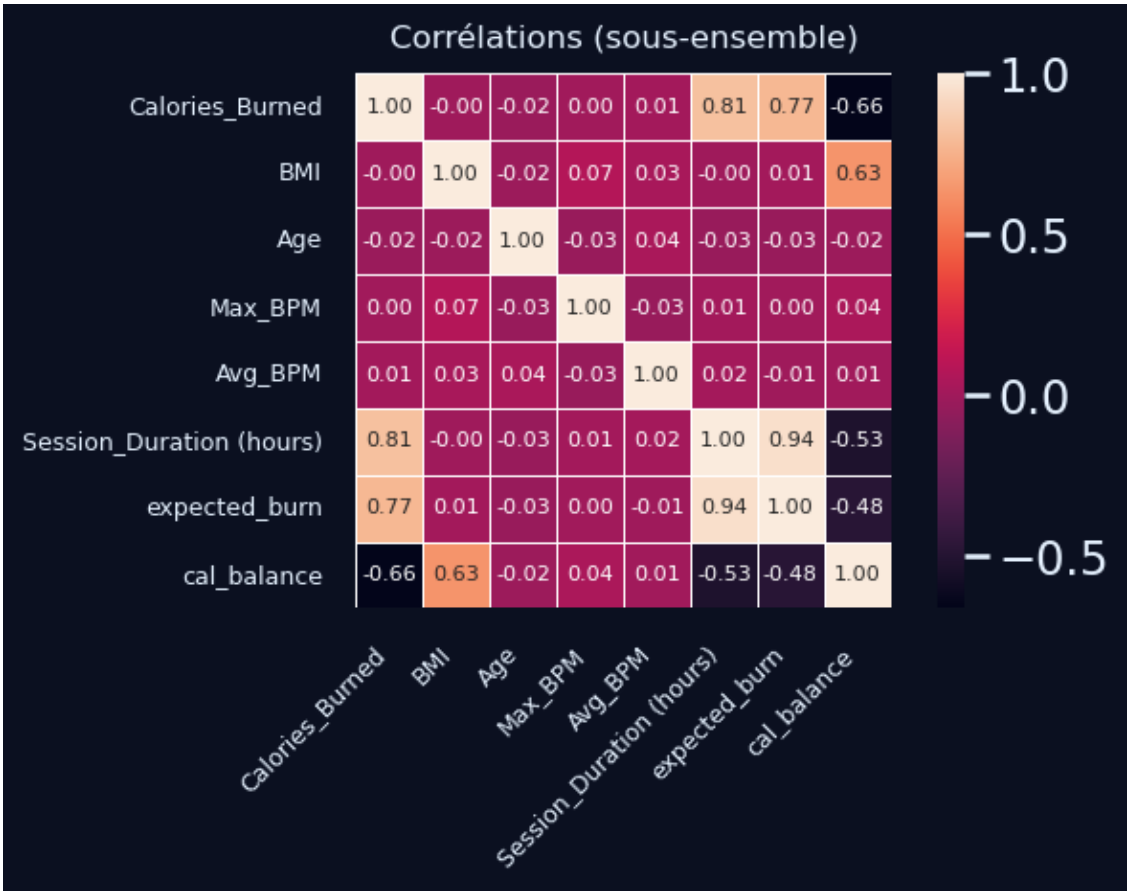
[74]:
```python
def iqr_outlier_ratio(s: pd.Series, k: float = 1.5) -> float:
    q1, q3 = s.quantile(0.25), s.quantile(0.75)
    iqr = q3 - q1
    if iqr == 0:
        return 0.0
```

```
    low, high = q1 - k*iqr, q3 + k*iqr
    return ((s < low) | (s > high)).mean() * 100

outlier_report = pd.Series({c: iqr_outlier_ratio(df[c]) for c in num_cols},␣
 ↪name="outliers_%")
display(outlier_report.sort_values(ascending=False).head(12))
```

```
Physical exercise              23.475
Burns Calories (per 30 min)     4.465
BMI                             2.650
BMI_calc                        2.645
Calories_Burned                 2.535
Carbs                           2.025
Proteins                        2.015
Fats                            1.995
cal_from_macros                 1.985
protein_per_kg                  1.635
pct_carbs                       1.535
cal_balance                     1.160
Name: outliers_%, dtype: float64
```

1.2.9 Aperçu corrélations (sous-ensemble lisible)

[76]:
```
subset_for_corr = [c for c in␣
 ↪["Calories_Burned","BMI","Age","Max_BPM","Avg_BPM",
                              "Session_Duration␣
 ↪(hours)","expected_burn","cal_balance"]
                  if c in num_cols]
corr = df[subset_for_corr].corr(numeric_only=True)
plt.figure(figsize=(7, 5))
sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    square=True,
    cbar=True,
    annot_kws={"size": 8},    #  taille des annotations
    linewidths=0.5,           # optionnel : lignes fines entre les cases
)

plt.title("Corrélations (sous-ensemble)", fontsize=12, pad=10)
plt.xticks(fontsize=9, rotation=45, ha="right")
plt.yticks(fontsize=9, rotation=0)
plt.tight_layout()
plt.show()
```



1.3 3.1.3. Visualisation univariée

1.3.1 Objectif

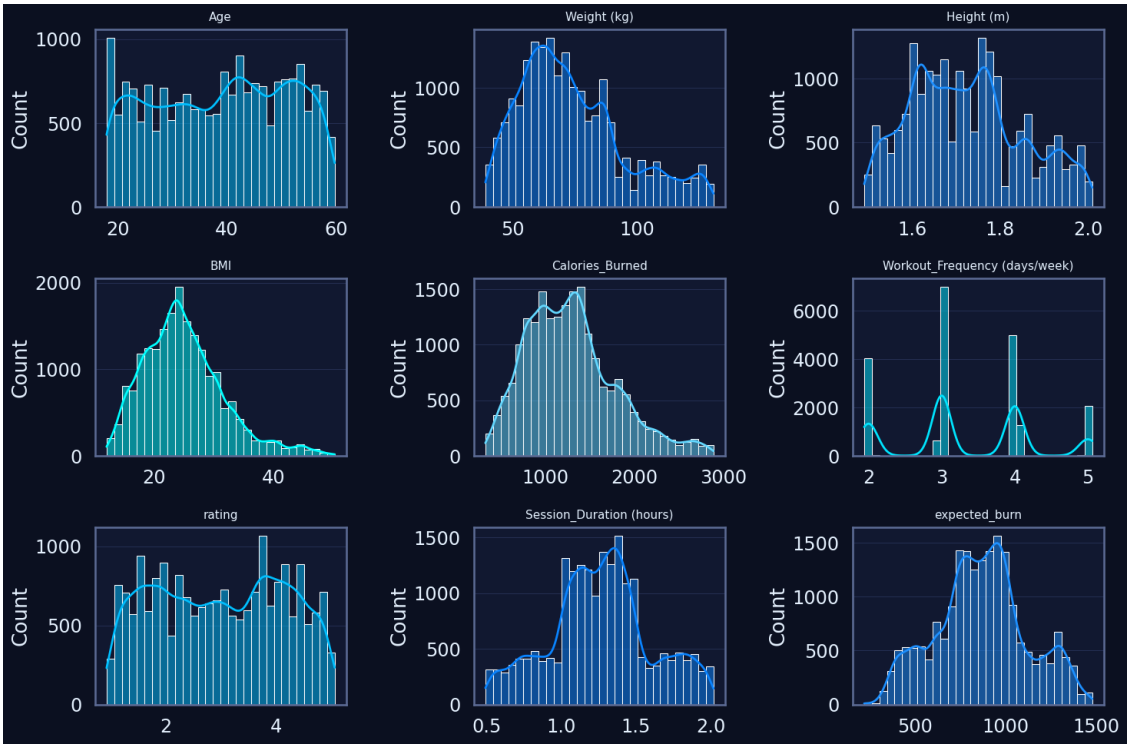
Analyser la distribution individuelle des principales variables numériques pour : - détecter les asymétries ou les concentrations atypiques, - identifier les profils dominants du dataset, - et valider la cohérence des mesures (âge, IMC, calories, fréquence d'entraînement…).

1.3.2 Interprétation attendue

- **Âge, poids et taille** : distributions normales, confirmant une population adulte équilibrée.
- **BMI et Calories_Burned** : distributions légèrement asymétriques vers la droite (quelques individus très actifs).
- **Workout_Frequency et Session_Duration** : centrées autour de valeurs moyennes (3 jours/semaine, 1,2 h/séance).
- **rating** : distribution presque uniforme, reflétant une évaluation variée des exercices.
- **expected_burn** : pic autour de la moyenne (~870 kcal), illustrant une population plutôt homogène en effort énergétique.

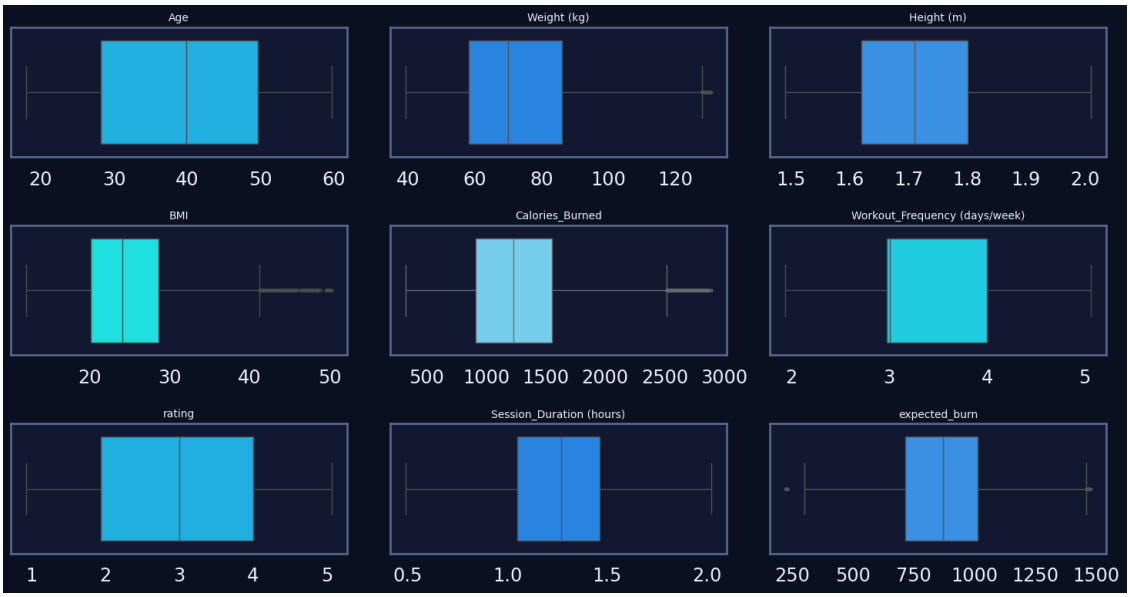
1.3.3 Visualisation des distributions univariées

[77]:
```
# Variables clés à visualiser
uni_vars = [
    "Age", "Weight (kg)", "Height (m)", "BMI",
    "Calories_Burned", "Workout_Frequency (days/week)",
    "rating", "Session_Duration (hours)", "expected_burn"
]

plt.figure(figsize=(15, 10))
for i, col in enumerate(uni_vars, 1):
    ax = plt.subplot(3, 3, i)
    sns.histplot(df[col], kde=True, color=palette_trainme[i %␣
 ↪len(palette_trainme)], bins=30)
    ax.set_title(col, fontsize=11, pad=8)
    ax.set_xlabel("")
plt.tight_layout()
plt.show()
```



1.3.4 Boxplots univariés pour détection visuelle d'outliers

[78]:
```
plt.figure(figsize=(15, 8))
for i, col in enumerate(uni_vars, 1):
    ax = plt.subplot(3, 3, i)
    sns.boxplot(x=df[col], color=palette_trainme[i % len(palette_trainme)],␣
 ↪fliersize=2)
    ax.set_title(col, fontsize=10)
    ax.set_xlabel("")
plt.tight_layout()
plt.show()
```



1.4 3.1.4. Visualisation bivariée

1.4.1 Objectif

Étudier les relations entre deux variables à la fois pour identifier : - des corrélations linéaires ou non linéaires, - des patterns comportementaux (ex. intensité calories brûlées), - des dépendances entre indicateurs physiologiques, nutritionnels et d'effort.

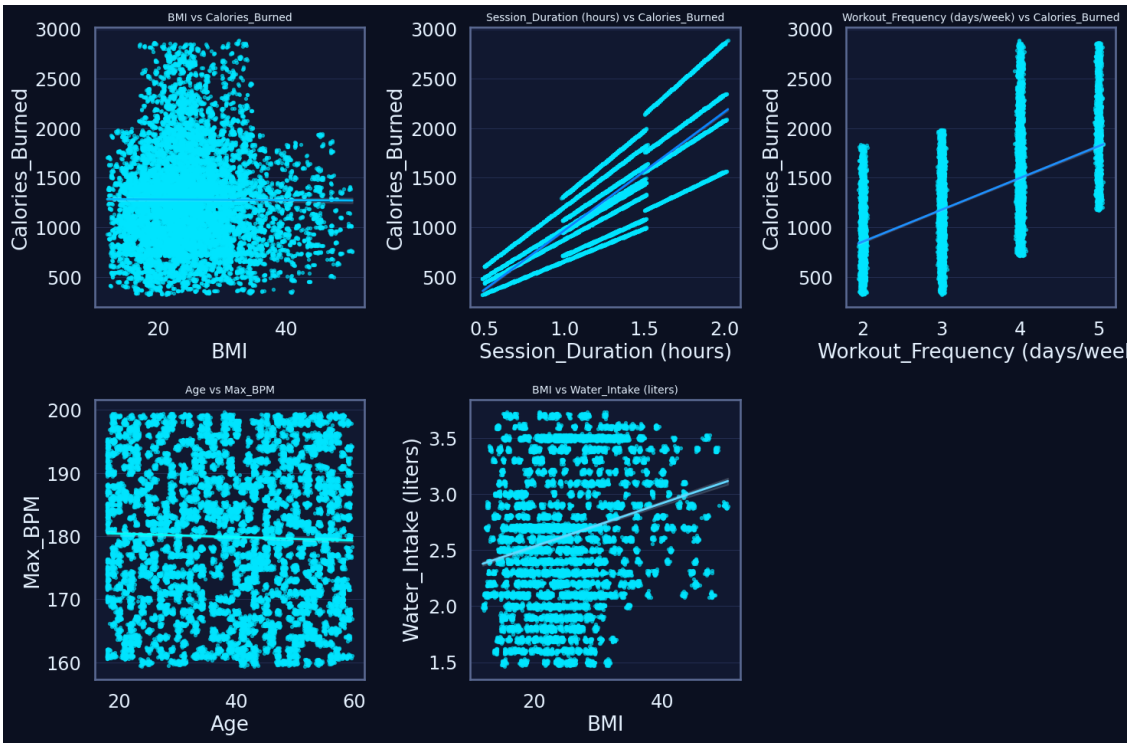
1.4.2 Interprétation attendue

- Relation positive nette entre **durée de séance** et **calories brûlées** : plus la séance est longue, plus la dépense énergétique augmente.
- L'**IMC** influence légèrement la dépense calorique, mais de façon dispersée : les individus plus massifs dépensent davantage, sans corrélation stricte.
- Corrélation inverse modérée entre **âge** et **fréquence cardiaque maximale**, conforme aux observations physiologiques.

- Les apports nutritionnels (calories, protéines, glucides) sont logiquement corrélés à la dépense énergétique.
- Ces relations confirment la cohérence physiologique du jeu de données et orientent les futures analyses multivariées.

1.4.3 Relations physiologiques et comportementales

[79]:
```
biv_pairs = [
    ("BMI", "Calories_Burned"),
    ("Session_Duration (hours)", "Calories_Burned"),
    ("Workout_Frequency (days/week)", "Calories_Burned"),
    ("Age", "Max_BPM"),
    ("BMI", "Water_Intake (liters)"),
]

plt.figure(figsize=(15, 10))
for i, (x, y) in enumerate(biv_pairs, 1):
    ax = plt.subplot(2, 3, i)
    sns.regplot(data=df, x=x, y=y, scatter_kws={"s": 8, "alpha": 0.6},
                line_kws={"color": palette_trainme[i % len(palette_trainme)],␣
 ↪"lw": 2})
    ax.set_title(f"{x} vs {y}", fontsize=10)
plt.tight_layout()
plt.show()
```



1.4.4 Corrélation nutrition performance

[80]:
```
biv_pairs2 = [
    ("Calories", "Calories_Burned"),
    ("Carbs", "expected_burn"),
    ("Proteins", "Calories_Burned"),
    ("Fats", "expected_burn"),
]

plt.figure(figsize=(15, 8))
for i, (x, y) in enumerate(biv_pairs2, 1):
    ax = plt.subplot(2, 2, i)
    sns.scatterplot(data=df, x=x, y=y,
                    color=palette_trainme[i % len(palette_trainme)], alpha=0.5,␣
 ↪s=15)
    sns.lineplot(data=df, x=x, y=y, color="white", lw=1.5, alpha=0.4)
    ax.set_title(f"{x} vs {y}", fontsize=10)
plt.tight_layout()
plt.show()
```